

HOMEWORK #3

Friday, December 5, 2025 1:17 PM

Q1 (10 points)

You toss a biased coin two times. The random variable, X , represents the number of Heads obtained. Therefore, X is a Binomial Random Variable with parameter $n = 2$ with a probability of success, $p = 0.2$.

a) Calculate and plot the PMF, $p_X(x)$ for the Discrete Random Variable X .

We now define a new discrete random variable Y , where the conditional PMF $P(Y = y|X = x) = p_{Y|X}(y|x)$ is a discrete uniform distribution taking on values of x to 4.

(For example, if $X = 2$, then $Y|X = 2$ has a discrete Uniform distribution on $\{2, 3, 4\}$).

b) Calculate and plot the conditional PMF $p_{Y|X}(y|x)$ for all values of x .

c) Calculate and plot the marginal PMF, $p_Y(y)$.

d) Calculate and plot the Joint PMF, $p_{X,Y}(x, y)$.

$$p = 0.2$$

$$X \sim \text{Bin}(n=2, p=0.2)$$

$$a) \quad P(X=x) = \binom{2}{x} (0.2)^x (0.8)^{2-x}, \text{ for } x=0,1,2$$

$$x=0$$

$$P(X=0) = \binom{2}{0} (0.2)^0 (0.8)^2 = \frac{16}{25}$$

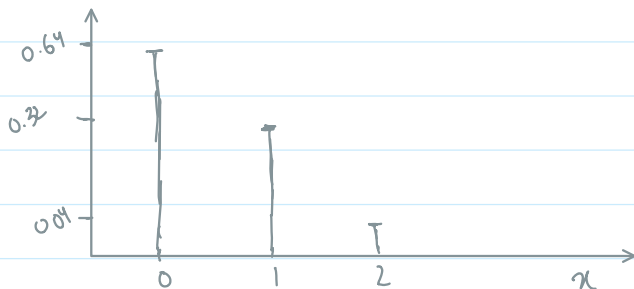
$$x=1:-$$

$$\binom{2}{1} (0.2)^1 (0.8)^1 = \frac{8}{25}$$

$$x=2:-$$

$$\binom{2}{2} (0.2)^2 (0.8)^0 = \frac{1}{25}$$

PMF:-



b) $X|X=x$ is discrete uniform at $[x, x+1, \dots, 4]$
 then y has equal probability.

$$P_{Y|X}(y|x) = \begin{cases} \frac{1}{5-x} & , y=x, x+1, \dots, 4 \\ 0 & , \text{otherwise} \end{cases}$$

$x=0$:

$\{0, 1, 2, 3, 4\} \rightarrow$ same prob.. that is ' $1/5$ '

y	0	1	2	3	4
	0	$1/5$	$1/5$	$1/5$	$1/5$

$x=1$:

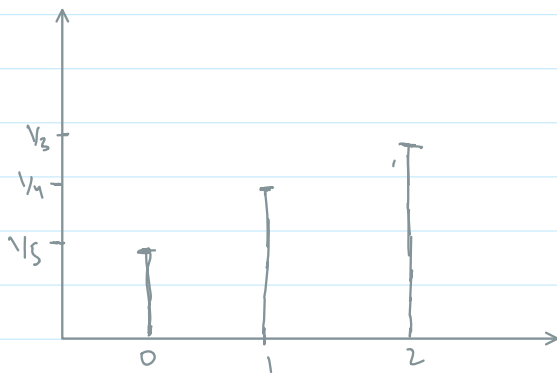
$\{1, 2, 3, 4\} \rightarrow 1/4 = p$

y	1	2	3	4
	0	$1/4$	$1/4$	$1/4$

$x=2$:

$\{1, 3, 4\} \rightarrow 1/3 = p$

y	0	1	2	3	4
	0	0	0	$1/3$	$1/3$



c) $Y=0$

$$P(Y=0) = P(Y=0 | X=0)P(X=0) = \frac{1}{5} \cdot \frac{16}{25} = \frac{16}{125}$$

$Y=1:-$

$$P(Y=1 | X=0) P(X=0) + P(Y=1 | X=1) P(X=1) \\ = \frac{1}{5} \cdot \frac{16}{125} + \frac{1}{4} \cdot \frac{8}{25} = \frac{26}{125}$$

$Y=2:-$

$$P(Y=2) = P(Y=2 | X=0) P(X=0) + P(Y=2 | X=1) P(X=1) + P(Y=2 | X=2) P(X=2) \\ = \frac{1}{5} \cdot \frac{16}{25} + \frac{1}{4} \cdot \frac{8}{25} + \frac{1}{3} \cdot \frac{1}{25} = \frac{83}{375}$$

Similarly for $Y=3:-$

$$P(Y=3 | X=0) = 1/5, \quad P(Y=3 | X=1) = 1/6, \quad P(Y=3 | X=2) = 1/7$$

$$P(Y=3) = \frac{48}{375} + \frac{30}{375} + \frac{5}{375} = \frac{83}{375}$$

$$P(Y=4) = \frac{83}{375}$$

d $P_{xy}(x, y) = P_{yx}(y | x) \cdot P_x(x)$

$$P_{x,y}(0,0) = 0.128$$

$$P_{x,y}(0,1) = 0.128$$

$$P(1,1) = 0.25 \times 0.32 = 0.08$$

$$P(2,1) = 1/3 \times 0.04 = 0.0133$$

Y	0		1	2	3	4
X	0	0.128	0.128	0.128	0.128	0.128
	1	0	0.06	0.08	0.08	0.08
	2	0	0	0.013	0.013	0.013

Q2 (10 points)

If the height of Habib University students is modeled as a Normal random variable with mean of 65 inches and standard deviation of 3 inches, what is the probability that a randomly chosen HU student has a height of 72 inches or more?

$$H \sim N(\mu = 65, \sigma = 3)$$

$$Z = \frac{H - \mu}{\sigma}$$

$$= \frac{72 - 65}{3} = \frac{7}{3}$$

$$Z = 2.33$$

$$P(H \geq 72) = P(Z \geq 2.33)$$

$$P(Z \leq 2.33) = 0.99$$

$$P(Z \geq 2.33) = 1 - 0.99 = 0.0099$$

Q3 (10 points)

The time until the next flight departs from an airport follows the following distribution:

$$f(x) = \frac{1}{20} \text{ for } 25 \leq x \leq 45$$

- What is $E[X]$?
- What is the standard deviation of X ?
- Find the probability that the time until the departure of the next flight is at most 30 minutes.
- Find the probability that the time until the departure of the next flight is between 30 and 40 minutes.

$$a) f(x) = \frac{1}{20} \quad 25 \leq x \leq 45$$

$$a = 25, \quad b = 45, \quad b - a = 20$$

$$E[X] = \frac{a+b}{2} = \frac{25+45}{2} = 35 \text{ min}$$

$$b) \text{var}(X) = \frac{(b-a)^2}{12}$$

$$SD(X) = \sqrt{\text{var}(X)} = \frac{b-a}{\sqrt{12}}$$

$$SD(x) = \sqrt{\text{var}(x)} = \frac{b-a}{\sqrt{12}}$$

$$\frac{20}{\sqrt{12}} = 5.77 \text{ min}$$

$$c) P(X \leq 30) = \int_{25}^{30} \frac{1}{20} dx = \frac{1}{4} = 0.25$$

$$d) P(30 \leq x \leq 40) = \int_{30}^{40} \frac{1}{20} dx = \frac{1}{2} = 0.5$$

Q4 (10 points)

Suppose that the number of kilometers you drive before you get into a car accident is modeled as an exponential random variable with a mean of 15,000 kms. You are planning to drive in your car from Karachi to Khunjerab Pass and back. What is the probability that you will make it back to Karachi without getting into an accident?

(Hint: According to Google Maps, the distance from Karachi to Khunjerab Pass is 2,100 km)

$$\mu = 15000$$

$$\text{Karachi} \longleftrightarrow \text{Khunjerab} = 2100 \times 2 = 4200 \text{ km}$$

$$E[X] = \frac{1}{\lambda}$$

$$15000 = \frac{1}{\lambda}$$

$$\lambda = \frac{1}{15000}$$

• $P(X > 4200) \rightarrow$ driving more than 4200 km without accident.

$$= e^{-1/15000 \times 4200}$$

$$= 0.7558$$

Q5 (10 points)

- a) Consider a javelin throw competition involving three athletes: Arshad, Babar, and Shaheen. Each athlete gets only one throw during the competition. Throws by each athlete are independent of the other athletes. Arshad can throw between 80m to 90m with all values being equally likely. Babar can throw between 85m to 90m with all values being equally likely. Shaheen can throw between 75m to 85m with all values being equally likely. Let random variable X be the distance (in meters) of the longest throw during the competition. What is the PDF of X ?
- b) X and Y are independent continuous random variables that are uniformly distributed with $X \sim U[0,2]$ and $Y \sim U[0,1]$. Let A be the event $\{0.25 \leq X \leq 0.5\}$. Find the following:
- $f_{X,Y}(1.5, 1.5)$
 - $F_{X,Y}(x, y)$
 - $E[X|A]$

- Babar $B \sim U[85, 90]$ $f_B(b) = \frac{1}{5}$, $85 \leq b \leq 90$
- Shaheens $S \sim U[75, 85]$ $f_S(s) = \frac{1}{10}$, $75 \leq s \leq 85$
- Arshad $A \sim U[80, 90]$ $f_A(a) = \frac{1}{10}$, $80 \leq a \leq 90$

Now,

$$P(A \leq x) = \frac{x-80}{10}, \quad P(B \leq x) = \frac{x-85}{5}, \quad P(S \leq x) = 1$$

$$F_x(x) = \frac{x-80}{10} \cdot \frac{x-85}{5} \cdot 1 = \frac{(x-80)(x-85)}{50}$$

So,

$$F_x(x) = \begin{cases} 0, & x < 85 \\ \frac{(x-80)(x-85)}{50}, & 85 \leq x \leq 90 \\ 1, & x > 90 \end{cases}$$

b)

i) $f_{X,Y}(x,y) = f_X(x) f_Y(y)$

X is uniform on $[0,1]$, so its PDF is 0 outside the interval

$y = 1.5 \notin [0,1]$, so,

$$f_Y(1.5) = 0$$

$$f_{X,Y}(1.5, 1.5) = 0$$

$$f_{x,y}(1.5, 1.5) = 0$$

$$ii) F_{x,y}(x,y) = F_x(x) F_y(y)$$

$$F_x(x) = \begin{cases} 0, & x < 0 \\ 1/2, & 0 \leq x \leq 2 \\ 1, & x > 2 \end{cases}$$

$$F_y(y) = \begin{cases} 0, & y < 0 \\ y, & 0 \leq y \leq 1 \\ 1, & y > 1 \end{cases}$$

Now,

$$i) P(A) = \int_{0.25}^{0.5} 0.5 \, dx = \frac{1}{8}$$

$$E[X|A] = \frac{1}{P(A)} \int_{0.25}^{0.5} x f_x(x) \, dx = \frac{1}{1/8} \int_{0.25}^{0.5} x \cdot \frac{1}{3} \, dx$$

$$\frac{1}{2} \left[\frac{x^2}{2} \right]_{0.25}^{0.5} = 0.04875 = \frac{0.04875}{0.125} = \frac{3}{8}$$

Q6 (10 points)

a) X and Y are random variables with a joint PDF given by:

$$f_{X,Y}(x,y) = \frac{xy^2}{81}$$

for $0 \leq x \leq 3$ and $0 \leq y \leq 3$. The joint PDF is zero for all other values of x and y . Are X and Y independent? Justify your answer.

b) X and Y are random variables with a joint PDF given by:

$$f_{X,Y}(x,y) = x + cy^3$$

for $0 \leq x \leq 1$ and $0 \leq y \leq 1$. The joint PDF is zero for all other values of x and y .

i) What is the value of constant c ?

ii) Are X and Y independent? Justify your answer.

$$a) f_{x,y}(x,y) = \frac{xy^2}{81}, \quad 0 \leq x \leq 3, \quad 0 \leq y \leq 3$$

Marginal of x :-

$$f_x(x) = \int_0^3 f_{x,y}(x,y) \, dy = \int_0^3 \frac{xy^2}{81} \, dy = \frac{x}{81} \left[\frac{y^3}{3} \right]_0^3$$

$$= \frac{x}{9} \quad 0 \leq x \leq 3$$

• marginal of y :-

$$f_y(y) = \int_0^3 \frac{xy^2}{81} dx = \frac{y^2}{81} \left[\frac{x^2}{2} \right]_0^3 = \frac{y^2}{18}$$

$0 \leq y \leq 3$

for independence:

$$f_x(x) \cdot f_y(y) = \frac{x}{9} \cdot \frac{y^2}{18} = \frac{xy^2}{162} \quad \text{as, } f_{xy}(x,y) = \frac{xy^2}{81}$$

so, they are not independent.

b)
$$\int_0^1 \int_0^1 (x + Cy^3) dy dx = 1$$

$$\int_0^1 (x + Cy^3) dy = x \int_0^1 dy + C \int_0^1 y^3 \cdot dy = 2(1) + C \left[\frac{y^4}{4} \right]_0^1$$

$$= \int_0^1 \left(x + \frac{C}{4} \right) dx = \frac{x^2}{2} \Big|_0^1 + \frac{C}{4} \cdot 1 = \frac{1}{2} + \frac{C}{4}$$

$$\frac{1}{2} + \frac{C}{4} = 1 \quad \text{so} \quad C = 2$$

ii)
$$f_x(x) = \int_0^1 (x + 2y^3) \cdot dy = x + \frac{1}{2}, \quad 0 \leq x \leq 1$$

$$f_y(y) = \int_0^1 (x + 2y^3) dx = \frac{1}{2} + 2y^3, \quad 0 \leq y \leq 1$$

$$f_{x,y}(x,y) = f_x(x) \cdot f_y(y) = \left(x + \frac{1}{2} \right) \left(\frac{1}{2} + 2y^3 \right)$$

$$= x + 2y^3$$

$$f_{x,y}(0,0) = 0$$

$$f_{X,Y}(0,0) = 0$$

So, they are not independent.

$$f_X(0) f_Y(0) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

Q7 (10 points)

X and Y are random variables with a joint PDF given by:

$$f_{X,Y}(x,y) = \begin{cases} \frac{2xy^2}{81} & 0 \leq x \leq 3, \quad 0 \leq y \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

Let A be the event $\{0 \leq X \leq 1\}$. Find the following:

i) $P(X \leq 1, Y \leq 1)$

ii) $E[X|A]$

i) $P(X \leq 1, Y \leq 1) \quad 0 \leq x \leq 1, \quad 0 \leq y \leq 1$

$$P(X \leq 1, Y \leq 1) = \int_0^1 \int_0^1 \frac{2xy^2}{81} dy dx$$

$$\int_0^1 \frac{2x}{81} y^2 dy = \frac{2x}{81} \int_0^1 y^2 dy = \frac{2x}{81} \left(\frac{y^3}{3} \right)_0^1 = \frac{2x}{243}$$

$$\int_0^1 \frac{2x}{243} dx = \frac{2}{243} \left(\frac{x^2}{2} \right)_0^1 = \frac{1}{243}$$

ii) $f_X(x) = \int_0^3 f_{X,Y}(x,y) dy = \int_0^3 \frac{2xy^2}{81} dy = \frac{2x}{81} \cdot \frac{y^3}{3} \Big|_0^3 = \frac{2x}{9}$

$$P(A) = P(0 \leq X \leq 1) = \int_0^1 \frac{2x}{9} dx = \frac{2}{9} \left[\frac{x^2}{2} \right]_0^1 = \frac{1}{9}$$

$$\int_A x f_X(x) dx = \int_0^1 x \cdot \frac{2x}{9} dx = \frac{2}{9} \left[\frac{x^3}{3} \right]_0^1 = \frac{2}{27}$$

$$E[X|A] = \frac{1}{P(A)} \int_A x f_X(x) dx = \frac{1}{1/9} \cdot \frac{2}{27} = \boxed{\frac{2}{3}}$$

$$E[X|A] = \frac{1}{P(A)} = \int_A x f_X(x) \cdot dx = 1 \cdot \frac{2}{27} = \left\lfloor \frac{2}{3} \right\rfloor$$

Q8 (10 points)

X and Y are independent continuous random variables that are uniformly distributed with $X \sim U[0, 2]$ and $Y \sim U[0, 2]$. Find the following:

- a) $f_{X,Y}(x, y)$
- b) $F_{X,Y}(1.5, 1.5)$
- c) $f_{Y|X}(y, 0.5)$
- d) $f_{X|Y}(0.5, 1)$
- e) $E[X|Y > 1.5]$

$$X \sim U[0, 2] \quad Y \sim U[0, 2]$$

$$f_X(x) = \frac{1}{2}, \quad 0 \leq x \leq 2 \quad f_Y(y) = \frac{1}{2}, \quad 0 \leq y \leq 2$$

$$a) \quad f_{X,Y}(x, y) = f_X(x) f_Y(y) = \begin{cases} \frac{1}{4}, & 0 \leq x \leq 2, \quad 0 \leq y \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

$$b) \quad F_X(x) = \frac{x}{2} \quad 0 \leq x \leq 2, \quad f_X(y) = \frac{y}{2}, \quad 0 \leq y \leq 2$$

$$F_{X,Y}(1.5, 1.5) = \frac{1.5}{2} \cdot \frac{1.5}{2} = (0.75)^2 = \frac{9}{16}$$

$$c) \quad f_{Y|X}(y|x) = \frac{f_{X,Y}(x, y)}{f_X(x)}$$

for $x = 0.5$:-

$$f_{Y|X}(y|0.5) = \frac{1/4}{1/2} = \frac{1}{2} \quad 0 \leq y \leq 2$$

$$d) \quad f_{X|Y}(x|y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}$$

$$\frac{1/4}{1/2} = \frac{1}{2}$$

$$e) \quad E[X|Y > 1.5] = E[X] = \frac{2}{2} = 1, \text{ it's uniform}$$

Q9 (10 points)

- a) Binary signal S is transmitted across a communication channel. The received signal is $Y = N + S$, where N is normal noise with zero mean and unit variance, independent of S . Let A be the event that the noise is between -0.25 and 0.25 (i.e. $N \in [-0.25, 0.25]$). What is $E[N|A]$? Justify your answer.
- b) Suppose you were told that the average odometer reading (number of kilometers a car has traveled in its lifetime so far) of all the red cars in Karachi was 55,000 km. Express this information in terms of two random variables X and Y . Define the random variables X and Y appropriately. (You can use any of the concepts applicable to two random variables such as joint PDF, conditional PDF, etc.)

a) Noise $N \sim N(0,1)$

$$E[N|A] = \frac{\int_{-0.25}^{0.25} n f_N(n) dn}{\int_{-0.25}^{0.25} f_N(n) \cdot dn}$$

as,

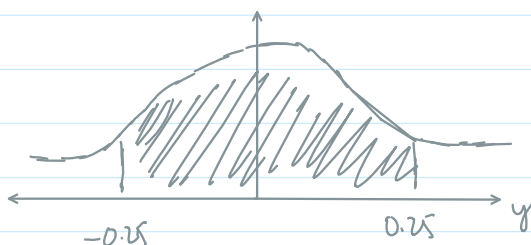
$$f_N(-n) = f_N(n)$$

$[-0.25, 0.25]$ is symmetric around zero '0'.

$$n f_N(n) = -(-n) f_N(-n)$$

so integral becomes zero.

$$\int_{-0.25}^{0.25} n f_N(n) dn = 0$$



$$E[N|A] = 0$$

b)

X = odometer reading

Y = indicator of one being red

$$Y = \begin{cases} 1, & \text{if chosen car is red} \\ 0, & \text{otherwise} \end{cases}$$

reading = 55000 so, $E[X|Y=1] = 55000 \text{ km}$

Q10 (10 points)

Consider a promising new batter who performed really well in the PSL this year. He has been called up to the training camp of Team Pakistan and is likely to start his career as an international player in the near future. Based on his batting technique, a cricket expert has provided you the following information about his potential career as a batter for Team Pakistan: By the end of his international career, his batting average (number of runs scored per innings) inside Asia will be between 30.5 to 40.5 runs with all values being equally likely. His batting average outside Asia will be between 15.5 and 35.5 with all values being equally likely. Team Pakistan plays 40% of its matches outside Asia. Let X be the continuous random variable that captures his batting average at the end of his international career with Team Pakistan, what is the PDF of X ?

for Asia,

$$f_{x_{in}}(x) = \frac{1}{40.5 - 30.5} = \frac{1}{10} \quad 30.5 \leq x \leq 40.5$$

for outside of Asia-

$$f_{x_{out}}(x) = \frac{1}{35.5 - 15.5} = \frac{1}{20} \quad 15.5 \leq x \leq 35.5$$

$$f_x(x) = P(in) f_{x_{in}}(x) + P(out) f_{x_{out}}(x)$$

1. $15.5 \leq x \leq 30.5$

$$f_x(x) = 0.4 \cdot \frac{1}{20} = 0.02$$

2. $30.5 \leq x \leq 35.5$

$$f_x(x) = 0.6 \cdot \frac{1}{10} + 0.4 \cdot \frac{1}{20} = 0.08$$

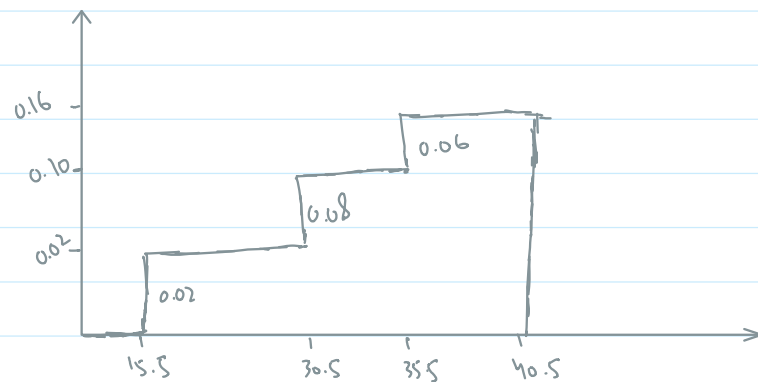
3. $35.5 \leq x \leq 40.5$

$$f_x(x) = 0.6 \cdot \frac{1}{10} = 0.06$$

outside $[15.5, 40.5]$,

so PDF is zero.

$$f_x(x) = \begin{cases} 0.02, & 15.5 \leq x \leq 30.5 \\ 0.08, & 30.5 \leq x \leq 35.5 \\ 0.06, & 35.5 \leq x \leq 40.5 \\ 0, & \text{otherwise} \end{cases}$$



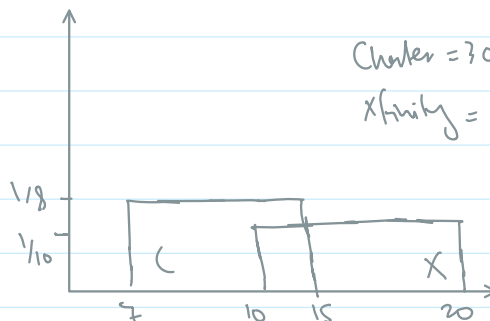
Q11 (10 points)

A city has two internet service providers, Charter and Xfinity, with the market share of 30% and 70% respectively. At any given time, the internet speed experienced by a customer of Charter is between 7Mbps and 15Mbps with all speeds being equally likely. At any given time, the internet speed experienced by a customer of Xfinity is between 10Mbps and 20Mbps with all speeds being equally likely. If an internet customer in the city is currently experiencing an internet speed of 12Mbps, what is the probability that the customer is served by Charter?

$$P(C|S=12)$$

$$f_{S|C}(s) = \begin{cases} \frac{1}{8} & , 7 \leq s \leq 15 \\ 0 & , \text{otherwise} \end{cases}$$

$$f_{S|X}(s) = \begin{cases} \frac{1}{10} & , 10 \leq s \leq 20 \\ 0 & , \text{otherwise} \end{cases}$$



Charter = 30%

Xfinity = 70%

Now,

$$P(C|S=12) = \frac{P(C) f_{S|C}(12)}{P(C) f_{S|C}(12) + P(X) f_{S|X}(12)} = \frac{0.3 \times \frac{1}{8}}{0.3 \times \frac{1}{8} + 0.7 \times \frac{1}{10}} = 0.348 \text{ Ans.}$$

Q12 (10 points)

A binary signal S is transmitted and we are given that $P(S=0) = 0.4$ and $P(S=1) = 0.6$. The received signal is $Y = N + S$, where N is normal noise with zero mean and unit variance, independent of S . What is the probability that $S=1$ if the observed value of received signal Y is 0.7.

for :-

$$Y|S=0 \sim N(0,1) \quad , \quad Y|S=1 \sim N(1,1)$$

$y=0.7$

$$\frac{1}{\sqrt{2\pi}} e^{-\frac{(y-\mu)^2}{2}}$$

$$f_{Y|S}(0.7|0) = \frac{1}{\sqrt{2\pi}} e^{-\frac{0.7^2}{2}} = \frac{1}{\sqrt{2\pi}} e^{-0.245}$$

So,

$$f_{Y|S}(0.7|1) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(0.7-1)^2}{2}}$$

• Acc to Bayes rule-

$$P(S=1 | Y=0.7) = \frac{P(S=1) f_{Y|S}(0.7 | 1)}{P(S=1) f_{Y|S}(0.7 | 1) + P(S=0) f_{Y|S}(0.7 | 0)}$$

$$P(S=1 | Y=0.7) = \frac{0.6}{0.6 + 0.4e^{-0.2}} = \underline{0.646} \text{ An}$$

Q13 (10 points)

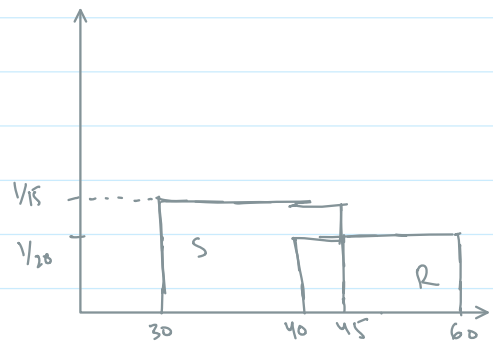
On a sunny day, it takes a student between 30 to 45 minutes to get to Habib University campus with all times being equally likely. On a rainy day, it takes the same student between 40 to 60 minutes to get to campus with all times being equally likely. Assume a day is sunny with probability 0.75 and rainy with probability 0.25.

- Find the PDF of the time T that it takes the student to get to campus.
- On a given day, it took the student 42 minutes to get to campus. What is the probability that this particular day was rainy?

$S \rightarrow \text{Sunny}, P(S) = 0.75$
 $R \rightarrow \text{Rainy}, P(R) = 0.25$

$$f_{T|S}(t) = \frac{1}{45-30} = \frac{1}{15}, \quad 30 \leq t \leq 45$$

$$f_{T|R}(t) = \frac{1}{60-40} = \frac{1}{20}, \quad 40 \leq t \leq 60$$



$$a) f_T(t) = P(S) f_{T|S}(t) + P(R) f_{T|R}(t)$$

$$1. \quad 30 \leq t \leq 40 \rightarrow f_T(t) = 0.75 \cdot \frac{1}{15} = 0.05$$

$$2. \quad 40 \leq t \leq 45 \rightarrow f_T(t) = 0.75 \cdot \frac{1}{15} + 0.25 \cdot \frac{1}{20} = 0.0625$$

$$3. \quad 45 \leq t \leq 60 \rightarrow f_T(t) = 0.25 \cdot \frac{1}{20} = 0.0125$$

$$f_T(t) = \begin{cases} 0.05, & 30 \leq t \leq 40 \\ 0.0625, & 40 \leq t \leq 45 \\ 0.0125, & 45 \leq t \leq 60 \\ 0, & \text{otherwise.} \end{cases}$$

b) Bayes'

$$P(R|T=42) = \frac{P(R) f_{T|R}(42)}{P(S) f_{T|S}(42) + P(R) f_{T|R}(42)}$$

for $t=42$, we are in the overlap $40 \leq t \leq 45$

$$f_{T|S}(42) = \frac{1}{15}, \quad f_{T|R}(42) = \frac{1}{20}$$

$$P(R|T=42) = \frac{0.25 \cdot 1/20}{0.75(1/15) + 0.25(1/20)} = 0.2$$

Q14 (10 points)

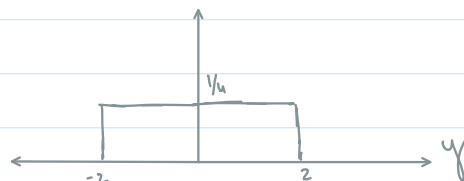
Consider a Bernoulli random variable X with parameter $p = 0.1$. We observe X in the presence of additive noise $N \sim U[-2, 2]$. As a result, our observation is random variable $Y = X + N$. If our observation is 0.8, what is the "most likely" value of X ?

(Note: Provide a justification of your answer based on the various theorems/rules/laws/results discussed in this course. Only intuitive argument is not acceptable for this question and will be given no partial credit)

$$f_{Y|X}(y|x)$$

$$Y = X + N \quad \& \quad N \sim U[-2, 2]$$

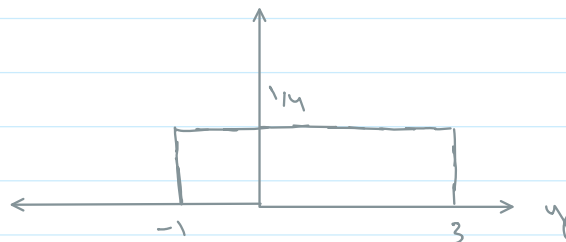
$$x=0 \rightarrow Y=N \rightarrow x=0 \sim U[-2, 2]$$



$$f_{Y|X}(y|0) = \begin{cases} 1/4, & -2 \leq y \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

$$x=1 \rightarrow Y=1+N \rightarrow x=1 \sim U[-1, 3]$$

$$f_{Y|X}(y|1) = \begin{cases} 1/4, & -1 \leq y \leq 3 \\ 0, & \text{otherwise} \end{cases}$$



$\therefore$ as $y=0.8$ lies in both intervals, $f_{Y|X}(0.8|0) = 1/4$

here $\wedge$

$$f_{Y|X}(0.8|0) = 1/4, \quad f_{Y|X}(0.8|1) = 1/4$$

$\bullet$ Bayes rule:-

$$P(X=x \mid Y=0.8) = \frac{P(X=x) f_{Y|X}(0.8|x)}{\sum_a P(X=a') f_{Y|X}(0.8|a')}$$

as, they take numerator and denominator are same,
so it becomes:-

$$P(X=x \mid Y=0.8) = P(X=x)$$

$$P(X=1 \mid Y=0.8) = 0.1, \quad P(X=0 \mid Y=0.8) = 0.9$$

Most likely $\boxed{X=0}$ Ans.